GITRINIL INJECTION

DESCRIPTION:
Gitrinil Injection 50mg/10ml: A clear, colourless solution

COMPOSITION:
Gitrinil Injection 50mg/10ml: Each ml contains Glycerin trinitrate 5mg. Each ampoule of 10ml contains Glyceryl trinitrate 50mg.

PHARMACODYNAMICS:
Vasodilating Agent. Glyceryl Trinitrate, an organic nitrate, is a vasodilator, its principal pharmacological action is the relaxation of vascular smooth muscle. Glyceryl Trinitrate produces, in a dose-related manner, dilation of both arterial and venous beds. Dilation of the post-capillary vessels, including large veins, promotes peripheral pooling of blood and decreases venous return to the heart, reducing left ventricular end-diastolic pressure (pre-load). Arteriolar relaxation reduces systemic vascular resistance and arterial pressure (after-load). Myocardial oxygen consumption or demand (as measured by the pressure-rate product, tension time index and stroke work index) is decreased by both the arterial and venous effects of glyceryl trinitrate, and a more favourable supply-demand ratio can be achieved.
Therapeutic doses of intravenous glyceryl trinitrate reduce systolic, diastolic and mean arterial blood pressure. Effective coronary perfusion pressure is usually maintained, but can be compromised if blood pressure falls extensively or increased heart rate decreases diastolic filling time.
Glyceryl trinitrate reduces elevated central venous and pulmonary capillary wedge pressures, pulmonary vascular resistance and systemic vascular resistance. Heart rate is usually slightly increased, presumably a reflex response to the fall in blood pressure. Cardiac index may be increased, decreased or unchanged. Patients with elevated left ventricular filling pressure and systemic vascular resistance values in conjunction with a depressed cardiac index are likely to experience an improvement in cardiac index. Alternatively, when filling pressures and cardiac index are normal, cardiac index may be slightly reduced by intravenous glyceryl trinitrate.

PHARMACOKINETICS:
Glyceryl Trinitrate is widely distributed in the body with an apparent volume of distribution of 200 L in adult male subjects. It is rapidly metabolised in the liver to dinitrates and mononitrates, with a short half-life estimated at 1-4 minutes. This results in a low plasma concentration after intravenous infusion. At plasma concentrations of between 50-500 ng/mL, the binding of glyceryl trinitrate to plasma proteins is approximately 60%, while those of its metabolites, 1,3-glyceryl dinitrate and 1,2-glyceryl dinitrate are approximately 60% and 30% respectively. The activity and half-life of dinitroglycerin metabolites are not well characterised. Glyceryl mononitrate, the principal metabolite, is inactive.

INDICATIONS:
Control of blood pressure in perioperative hypertension, i.e. hypertension associated with surgical procedures, especially cardiovascular procedures, such as the hypertension seen during intratracheal intubation, anaesthesia, skin incision, sternotomy, cardiac bypass and in the immediate post- surgical period.
Congestive heart failure associated with acute myocardial infarction.
Treatment of angina pectoris in patients who have not responded to recommended doses of organic nitrates and/or a beta-blocker. Production of controlled hypotension during neurosurgical and orthopaedic surgical procedures.

ROUTE OF ADMINISTRATION: For I.V. infusion only.

RECOMMENDED DOSAGE:
Dosage and Administration: NOT FOR DIRECT INTRAVENOUS INJECTION.
Glyceryl trinitrate injection is a concentrated potent drug which must be diluted in Dextrose BP or Sodium Chloride 0.9% injection BP prior to its infusion.
Glyceryl trinitrate injection should not be admixed with other drugs.
Initial dilution: To obtain a final concentration of 100 mcg/mL glyceryl trinitrate, aseptically transfer the contents of one Glyceryl trinitrate injection 50 mg/10 ml ampoule into a 500 mL glass bottle of either Dextrose injection BP or Sodium Chloride 0.9% injection BP.
Maintenance dilution: It is important to consider the fluid requirements of the patient as well as the expected duration of infusion in selecting the appropriate dilution of glyceryl trinitrate injection. After initial dosage titration, the concentration of the admixture solution may be increased, if necessary, to limit fluids given to the patient.
The glyceryl trinitrate concentration should not exceed 400 mcg/mL. Invert the glass parenteral bottle several times following admixture to assure uniform dilution of glyceryl trinitrate.
When stored in glass containers, the diluted solution is physically and chemically stable for up to 40 hours when stored below 25°C and up to seven days under refrigeration. However, the infusion should be commenced as soon as is practicable after preparation to avoid microbial contamination. Dosage is affected by the type of containers and administration sets used. *See Warnings*.
Although the usual starting adult dose range reported in clinical studies was 25 mcg/min or more, these studies used PVC administration sets. Doses need to be reduced with the use of non-absorbing tubing. The dosage of glyceryl trinitrate when non-absorbing tubing is used should initially be 5 mcg/min delivered through an infusion pump capable of exact and constant delivery of the drug. Subsequent titration must be adjusted to the clinical situation, with dose increments becoming more cautious as partial response is seen. Initial titration should be in 5 mcg/min increments, with increases every 3-5 minutes until some response is noted. If no response is seen at 20 mcg/min, increments of 10 and later 20 mcg/min can be used. Once a partial blood pressure response is observed, the dose increase should be reduced and the interval between increases should be lengthened.
Some patients with normal or low left ventricular filling pressures or pulmonary capillary wedge pressure (e.g. angina patients without other complications) may be hypersensitive to the effects of glyceryl trinitrate and may respond fully to doses as small as 5 mcg/min. These patients require especially careful titration and monitoring.
There is not fixed optimum dose of glyceryl trinitrate. Due to variations in the responsiveness of individual patients to the drug each patients must be titrated to the desired level of haemodynamic function. Therefore, continuous monitoring of physiologic parameters (e.g. blood pressure, heart rate, and pulmonary capillary wedge pressure) MUST BE PERFORMED to achieve the correct dose. Adequate systemic blood pressure and coronary perfusion pressure must be maintained.

CONTRAINDICATIONS:
A known hypersensitivity to glyceryl trinitrate or a known idiosyncratic reaction to organic nitrates.
Hypotension or uncorrected hypovolaemia, as the use of Glyceryl trinitrate, in such states, could produce severe hypotension or shock.
Increased intracranial pressure (e.g. head trauma or cerebral haemorrhage).
Constrictive pericarditis and pericardial tamponade.
Severe anaemia and arterial hypoxaemia.
Hypersensitivity to propylene glycol.

WARNINGS & PRECAUTIONS:

WARNING: NOT FOR DIRECT INTRAVENOUS INJECTION

Glyceryl trinitrate injection must be diluted in Glucose 5% injection BP or Sodium Chloride 0.9% injection BP, prior to infusion (see *Dosage and Administration)*. The administration set used will affect the amount of glyceryl trinitrate delivered to the patient (see *Warnings and Dosage and Administration*).
Glyceryl trinitrate readily migrates into many plastics. To avoid absorption of glyceryl trinitrate into plastic parenteral solution containers, the dilution and storage of glyceryl trinitrate for intravenous infusion should be made only in glass parenteral solution bottles.
Some filters absorb glyceryl trinitrate; they should be avoided. Forty to eighty percent of the total amount of glyceryl trinitrate in the final diluted solution for infusion is absorbed by the polyvinyl chloride (PVC) tubing of the intravenous administration sets currently in general use. The higher rates of absorption occur when flow rates are low, glyceryl trinitrate concentrations are high, and the administration set is long.
Although the rate of loss is highest during the early phase of infusion (when flow rates are lowest), the loss is neither constant nor self-limiting; consequently no simple calculation or correction can be performed to convert the theoretical infusion rate (based on the concentration of the infusion solution) to the actual delivery rate.
Acute myocardial infarction: Glyceryl trinitrate should only be used in acute myocardial infarction for treating definite left ventricular failure. Careful haemodynamic monitoring must be observed during infusion of glyceryl trinitrate in patients with myocardial infarction to avoid a sudden fall in arterial blood pressure and reflex tachycardia which might, by reducing coronary perfusion and increasing myocardial oxygen demand, extend the area of ischaemic tissue injury.
Hypotension: Dosage must be carefully titrated to avoid the significant risk of precipitous fall in blood pressure, particularly in patients with severe coronary or cerebral atherosclerosis. Profound hypotension and bradycardia have been reported with sublingual, topical and intravenous glyceryl trinitrate.
Excessive hypotension, especially for prolonged periods of time, must be avoided because of possible deleterious effects on the brain, heart, liver, and kidney from impaired perfusion and the attendant risk of ischaemia, thrombosis and altered functions of these organs. The use of vasodilators in hypertensive patients has been suspected of causing acute blindness. If pulmonary capillary wedge pressure is being monitored, it will be noted that a fall in wedge pressure precedes the onset of arterial hypotension and the pulmonary capillary wedge pressure is thus a useful guide to safe titration of the drug.
Precautions: Glyceryl trinitrate should be used with caution in patients with severe liver or renal disease. Paradoxical bradycardia and increased angina pectoris may accompany glyceryl trinitrate-induced hypotension. Long-term administration of organic nitrates may induce tolerance or cross-tolerance to glyceryl trinitrate or other organic nitrates in some patients, but tolerance does not appear to be clinically significant. Larger doses of glyceryl trinitrate may be required with chronic sublingual administration or when a patient is receiving oral nitrate vasodilators. Although clinical tolerance has not been observed during intravenous infusion of glyceryl trinitrate, the possibility of this occurring should be kept in mind. Nitrate dependence is a potentially serious problem. Death, myocardiac infarction, coronary spasm and chest pain syndromes have been documented in industrial workers who leave the work environment for several days after exposure to glyceryl trinitrate and nitroglycol. There is some clinical evidence of

nitrate dependence in patients with both angina and congestive heart failure. Although withdrawal syndromes have not been reported to occur following IV infusion of glyceryl trinitrate for up to 9 days, it may be necessary to carefully taper therapy in patients with proven coronary atherosclerosis who have received prolonged high-dose infusions of the drug.
Arterial oxygen tension decreases after administration of glyceryl trinitrate in normal subjects and in patients with coronary artery disease. Hypoxaemia occurs as a result of increased pulmonary ventilation perfusion mismatch; however, the clinical significance of this effect is unknown. Caution should be observed in patients with severe ischaemic heart disease as a decrease in available oxygen may oppose its antianginal effect.
Methaemoglobinaemia has been reported in association with glyceryl trinitrate therapy. Methaemoglobinaemia may be clinically significant, especially in the presence of methaemoglobin reductase deficiencies or in congenital M haemoglobulin variants.
Use in children: The use of glyceryl trinitrate in children is not recommended, as its safety and effectiveness in children have not been established.

INTERACTION WITH OTHER MEDICATIONS:

Alcohol: Concomitant use with alcohol may cause hypotension due to an enhanced vasodilatory effect of glyceryl trinitrate.
Other drugs: Information on potential interaction with other drugs is poorly documented. Caution should be observed if other drugs are given concomitantly during infusion or glyceryl trinitrate as an interaction may adversely affect the haemodynamic response to the drug. Careful haemodynamic monitoring is essential.
Pancuronium: Infusion of glyceryl trinitrate increases the duration of pancuronium-induced neuromuscular blockade. This clinical observation has been reported by studies in the cat, however, neuromuscular blockade induced by succinylcholine and d-tubocurarine were not prolonged by glyceryl trinitrate. The mechanism of the interaction is unknown.
Morphine: One patient who received four 3 mg doses of I.V. morphine sulphate during a 24 hour period with IV glyceryl trinitrate, became unarousable, eventually responding to nalorphine. The possibility that IV glyceryl trinitrate slows morphine metabolism in this patient has been suggested. Caution should be observed when administering these two drugs simultaneously.
Tricyclic Antidepressants: Anticholinergic Agents. Caution should be observed when giving tricyclic anti-depressants and anticholinergic agents concomitantly with IV glyceryl trinitrate because these agents may potentiate the hypotensive effect of glyceryl trinitrate.

USE IN PREGNANCY & LACTATION:

Use in pregnancy: Animal reproductive studies have not been observed. Glyceryl trinitrate should only be given to a pregnant woman if clearly needed.
Use in lactation: It is not known whether glyceryl trinitrate is excreted in human milk. Caution should be exercised if there is a need to administer glyceryl trinitrate to a nursing woman.

SIDE EFFECTS:

The most frequent adverse reaction in patients treated with glyceryl trinitrate is headache, which occurs in approximately 2% of patients. Other adverse reactions occurring in less than 1% of patients are the following: tachycardia, nausea, vomiting, apprehension, restlessness, muscle twitching, retrosternal discomfort, palpitations, dizziness and abdominal pain.
The following additional adverse reactions have been reported with oral and/or topical use of glyceryl trinitrate: cutaneous flushing, weakness, and occasionally drug rash or exfoliative dermatitis, hypertension, bradycardia, methaemoglobinaemia, postural hypotension and syncope on assuming upright posture and withdrawal syndrome, e.g. increased frequency of angina attack.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Overdosage: Over dosage of glyceryl trinitrate may result in severe hypotension, transient headache and reflex tachycardia.
Treatment: Patients should be treated with elevating the legs and decreasing or temporarily terminating the infusion until the patient's condition stabilises. Since the duration of the haemodynamic effects following glyceryl trinitrate administration is quite short, additional corrective measures are usually not required. However, if further therapy is indicated, administration of an intravenous alpha adrenergic agonist (e.g. dopamine) should be considered.
Methaemoglobinaemia has been reported in accidental overdosage. It can be treated with Methylene Blue 1 mg/kg bodyweight given by slow IV infusion.

INCOMPATIBILITIES:

Glyceryl Trinitrate is not compatible with polyvinylchloride (PVC) and severe losses of glyceryl trinitrate (up to 50%) may occur if polyvinylchloride is used, resulting in a reduction of delivered dose and efficacy. Contact of the solution with polyvinylchloride bags should be avoided.
The product is compatible with glass infusion sets and with rigid infusion packs made of polyethylene; it may also be infused slowly using a syringe pump with a glass or plastic syringe.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Keep out of reach of children. *Jauhkan daripada kanak-kanak.*

PACK SIZE:

Packs of 5 ampoules x 10 ml.

SHELF LIFE:

Please refer to outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD (42491 M)
Lot 2599 Jalan Seruling 59, Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.

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